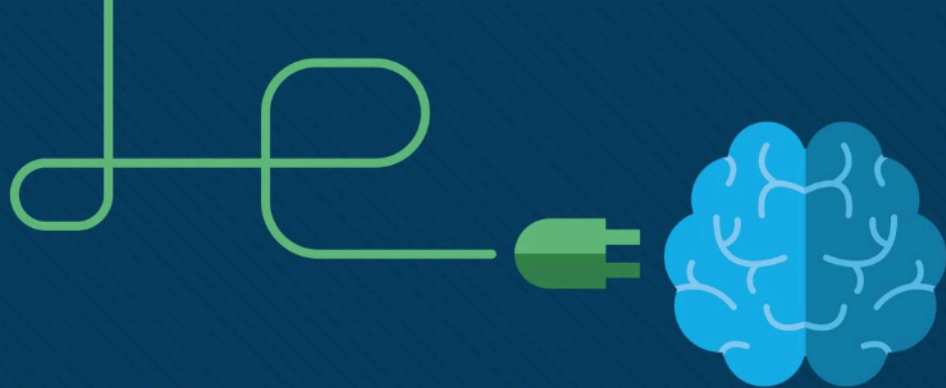




Module 7: DHCPv4

Switching, Routing and Wireless
Essentials v7.0 (SRWE)



Module Objectives

Module Title: DHCPv4

Module Objective: Implement DHCPv4 to operate across multiple LANs

Topic Title	Topic Objective
DHCP4 Concepts	Explain how DHCPv4 operates in a small- to medium-sized business network.
Configure a Cisco IOS DHCP4 Server	Configure a router as a DHCPv4 server.
Configure a DHCP4 Client	Configure a router as a DHCPv4 client.

7.1 DHCPv4 Concepts

DHCPv4 Server and Client

- Dynamic Host Configuration Protocol v4 (DHCPv4) assigns IPv4 addresses and other network configuration information dynamically. Because desktop clients typically make up the bulk of network nodes, DHCPv4 is an extremely useful and timesaving tool for network administrators.
- A dedicated DHCPv4 server is scalable and relatively easy to manage. However, in a small branch or SOHO location, a Cisco router can be configured to provide DHCPv4 services without the need for a dedicated server. Cisco IOS software supports an optional, full-featured DHCPv4 server.
- The DHCPv4 server dynamically assigns, or leases, an IPv4 address from a pool of addresses for a limited period of time chosen by the server, or until the client no longer needs the address.
- Clients lease the information from the server for an administratively defined period. Administrators configure DHCPv4 servers to set the leases to time out at different intervals. The lease is typically anywhere from 24 hours to a week or more. When the lease expires, the client must ask for another address, although the client is typically reassigned the same address.

DHCPv4 Concepts

DHCPv4 Operation

DHCPv4 works in a client/server mode. When a client communicates with a DHCPv4 server, the server assigns or leases an IPv4 address to that client.

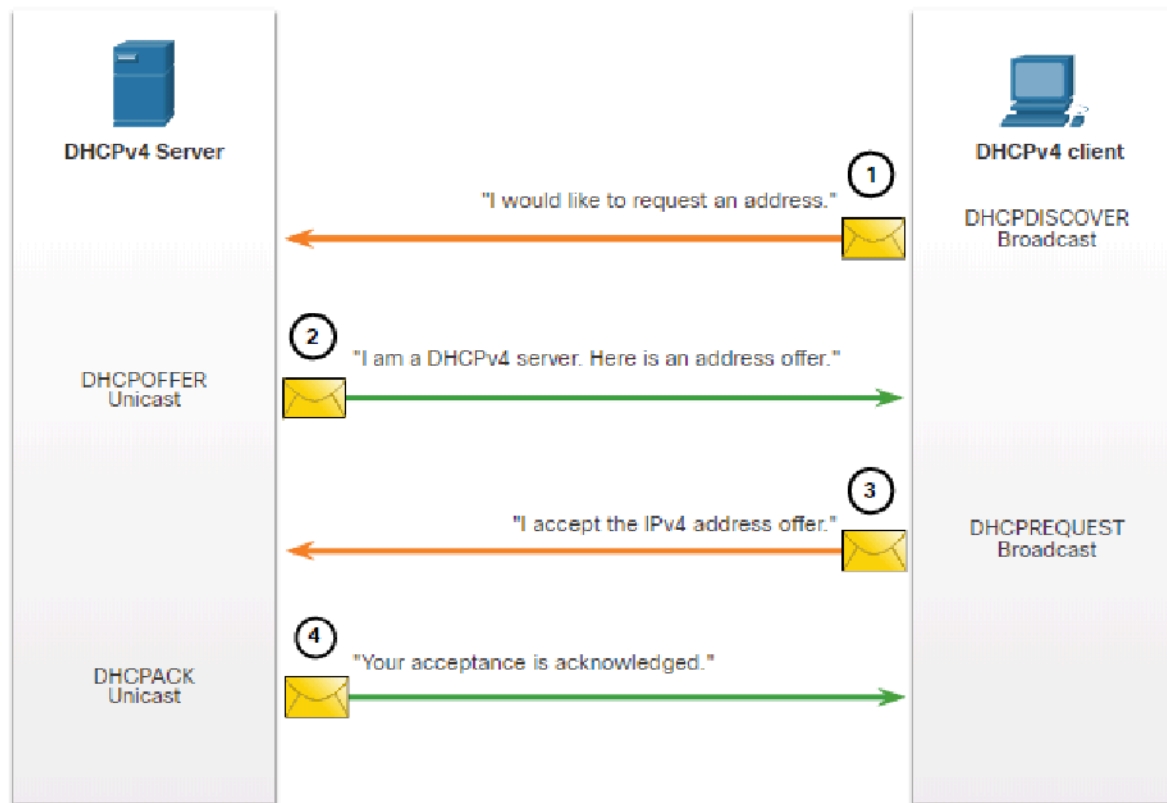
- The client connects to the network with that leased IPv4 address until the lease expires. The client must contact the DHCP server periodically to extend the lease.
- This lease mechanism ensures that clients that move or power off do not keep addresses that they no longer need.
- When a lease expires, the DHCP server returns the address to the pool where it can be reallocated as necessary.

DHCPv4 Concepts

Steps to Obtain a Lease

When the client boots (or otherwise wants to join a network), it begins a four-step process to obtain a lease:

1. DHCP Discover (DHCPDISCOVER)
2. DHCP Offer (DHCPOFFER)
3. DHCP Request (DHCPREQUEST)
4. DHCP Acknowledgment (DHCPACK)



DHCPv4 Concepts

Steps to Renew a Lease

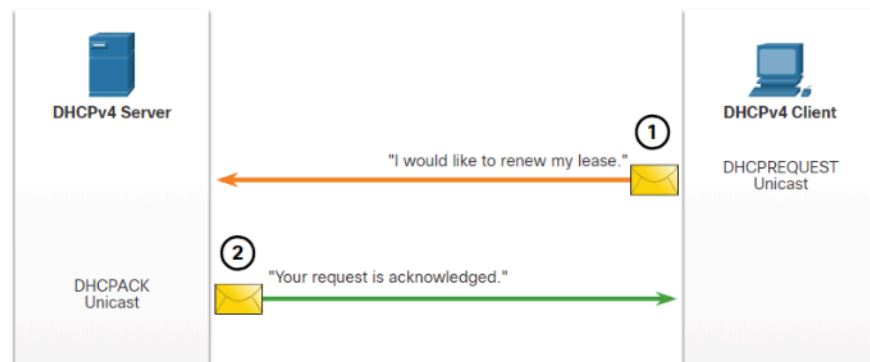
Prior to lease expiration, the client begins a two-step process to renew the lease with the DHCPv4 server, as shown in the figure:

1. DHCP Request (DHCPREQUEST)

Before the lease expires, the client sends a DHCPREQUEST message directly to the DHCPv4 server that originally offered the IPv4 address. If a DHCPACK is not received within a specified amount of time, the client broadcasts another DHCPREQUEST so that one of the other DHCPv4 servers can extend the lease.

2. DHCP Acknowledgment (DHCPACK)

On receiving the DHCPREQUEST message, the server verifies the lease information by returning a DHCPACK.



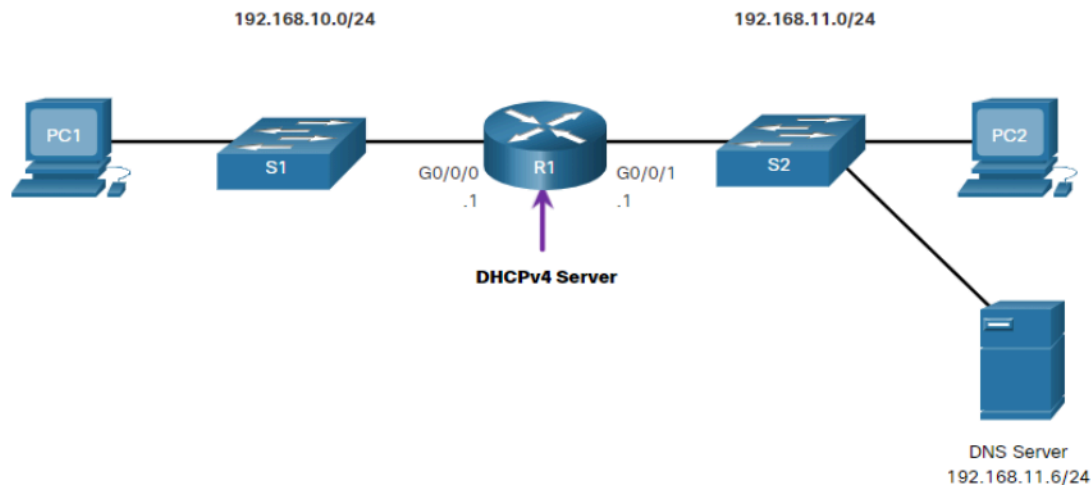
Note: These messages (primarily the DHCP OFFER and DHCPACK) can be sent as unicast or broadcast according to IETF RFC 2131.

7.2 Configure a Cisco IOS DHCPv4 Server

Configure a Cisco IOS DHCPv4 Server

Cisco IOS DHCPv4 Server

Now you have a basic understanding of how DHCPv4 works and how it can make your job a bit easier. A Cisco router running Cisco IOS software can be configured to act as a DHCPv4 server. The Cisco IOS DHCPv4 server assigns and manages IPv4 addresses from specified address pools within the router to DHCPv4 clients.



Steps to Configure a Cisco IOS DHCPv4 Server

Use the following steps to configure a Cisco IOS DHCPv4 server:

- **Step 1.** Exclude IPv4 addresses. A single address or a range of addresses can be excluded by specifying the *low-address* and *high-address* of the range. Excluded addresses should be those addresses that are assigned to routers, servers, printers, and other devices that have been, or will be, manually configured. You can also enter the command multiple times. The command is **ip dhcp excluded-address *low-address* [*high-address*]**
- **Step 2.** Define a DHCPv4 pool name. The **ip dhcp pool *pool-name*** command creates a pool with the specified name and puts the router in DHCPv4 configuration mode, which is identified by the prompt **Router(dhcp-config)#**.

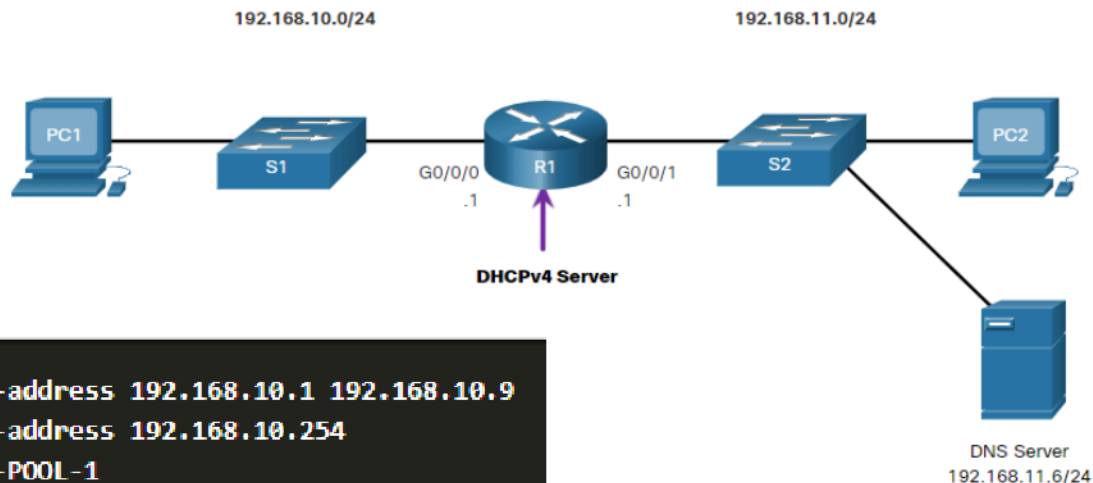
Steps to Configure a Cisco IOS DHCPv4 Server (Cont.)

- **Step 3.** Configure the DHCPv4 pool. The address pool and default gateway router must be configured. Use the **network** statement to define the range of available addresses. Use the **default-router** command to define the default gateway router. These commands and other optional commands are shown in the table.

Task	IOS Command
Define the address pool.	network <i>network-number</i> [<i>mask</i> / <i>prefix-length</i>]
Define the default router or gateway.	default-router <i>address</i> [<i>address2</i> <i>address8</i>]
Define a DNS server.	dns-server <i>address</i> [<i>address2</i> ... <i>address8</i>]
Define the domain name.	domain-name <i>domain</i>
Define the duration of the DHCP lease.	lease { <i>days</i> [<i>hours</i> [<i>minutes</i>]] infinite }
Define the NetBIOS WINS server.	netbios-name-server <i>address</i> [<i>address2</i> ... <i>address8</i>]

Configure a Cisco IOS DHCPv4 Server

Configuration Example



```
R1(config)# ip dhcp excluded-address 192.168.10.1 192.168.10.9
R1(config)# ip dhcp excluded-address 192.168.10.254
R1(config)# ip dhcp pool LAN-POOL-1
R1(dhcp-config)# network 192.168.10.0 255.255.255.0
R1(dhcp-config)# default-router 192.168.10.1
R1(dhcp-config)# dns-server 192.168.11.5
R1(dhcp-config)# domain-name example.com
R1(dhcp-config)# end
R1#
```

Configure a Cisco IOS DHCPv4 Server

DHCPv4 Verification

Use the commands in the table to verify that the Cisco IOS DHCPv4 server is operational.

Command	Description
show running-config section dhcp	Displays the DHCPv4 commands configured on the router.
show ip dhcp binding	Displays a list of all IPv4 address to MAC address bindings provided by the DHCPv4 service.
show ip dhcp server statistics	Displays count information regarding the number of DHCPv4 messages that have been sent and received

Configure a Cisco IOS DHCPv4 Server

Verify DHCPv4 is Operational

Verify the DHCPv4 Configuration: As shown in the example, the **show running-config | section dhcp** command output displays the DHCPv4 commands configured on R1. The **| section** parameter displays only the commands associated with DHCPv4 configuration.

```
R1# show running-config | section dhcp
ip dhcp excluded-address 192.168.10.1 192.168.10.9
ip dhcp excluded-address 192.168.10.254
ip dhcp pool LAN-POOL-1
  network 192.168.10.0 255.255.255.0
  default-router 192.168.10.1
  dns-server 192.168.11.5
  domain-name example.com
```

Configure a Cisco IOS DHCPv4 Server

Verify DHCPv4 is Operational (Cont.)

Verify DHCPv4 Bindings: As shown in the example, the operation of DHCPv4 can be verified using the **show ip dhcp binding** command. This command displays a list of all IPv4 address to MAC address bindings that have been provided by the DHCPv4 service.

```
R1# show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration        Type      State      Interface
                Hardware address/
                User name
192.168.10.10    0100.5056.b3ed.d8  Sep 15 2019 8:42 AM    Automatic Active
GigabitEthernet0/0/0
```

Configure a Cisco IOS DHCPv4 Server

Verify DHCPv4 is Operational (Cont.)

Verify DHCPv4 Statistics: The output of the **show ip dhcp server statistics** is used to verify that messages are being received or sent by the router. This command displays count information regarding the number of DHCPv4 messages that have been sent and received.

```
R1# show ip dhcp server statistics
Memory usage           19465
Address pools          1
Database agents        0
Automatic bindings     2
Manual bindings        0
Expired bindings       0
Malformed messages     0
Secure arp entries     0
Renew messages         0
Workspace timeouts     0
Static routes          0
Relay bindings         0
Relay bindings active   0
Relay bindings terminated 0
Relay bindings selecting 0
Message                Received
BOOTREQUEST            0
DHCPDISCOVER           4
DHCPREQUEST            2
DHCPDECLINE            0
DHCPRELEASE            0
DHCPIFORM              0
```


Configure a Cisco IOS DHCPv4 Server

Verify DHCPv4 is Operational (Cont.)

Verify DHCPv4 Client Received IPv4

Addressing: The **ipconfig**

/all command, when issued on PC1, displays the TCP/IP parameters, as shown in the example. Because PC1 was connected to the network segment 192.168.10.0/24, it automatically received a DNS suffix, IPv4 address, subnet mask, default gateway, and DNS server address from that pool. No DHCP-specific router interface configuration is required. If a PC is connected to a network segment that has a DHCPv4 pool available, the PC can obtain an IPv4 address from the appropriate pool automatically.

```
C:\Users\Student> ipconfig /all
Windows IP Configuration

Host Name . . . . . : ciscolab
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : example.com
    Description . . . . . : Realtek PCIe GBE Family Controller
    Physical Address. . . . . : 00-05-9A-3C-7A-00
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    IPv4 Address. . . . . : 192.168.10.10
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained . . . . . : Saturday, September 14, 2019 8:42:22AM
    Lease Expires . . . . . : Sunday, September 15, 2019 8:42:22AM
    Default Gateway . . . . . : 192.168.10.1
    DHCP Server . . . . . : 192.168.10.1
    DNS Servers . . . . . : 192.168.11.5
```

Configure a Cisco IOS DHCPv4 Server

Disable the Cisco IOS DHCPv4 Server

The DHCPv4 service is enabled by default. To disable the service, use the **no service dhcp** global configuration mode command. Use the **service dhcp** global configuration mode command to re-enable the DHCPv4 server process, as shown in the example. Enabling the service has no effect if the parameters are not configured.

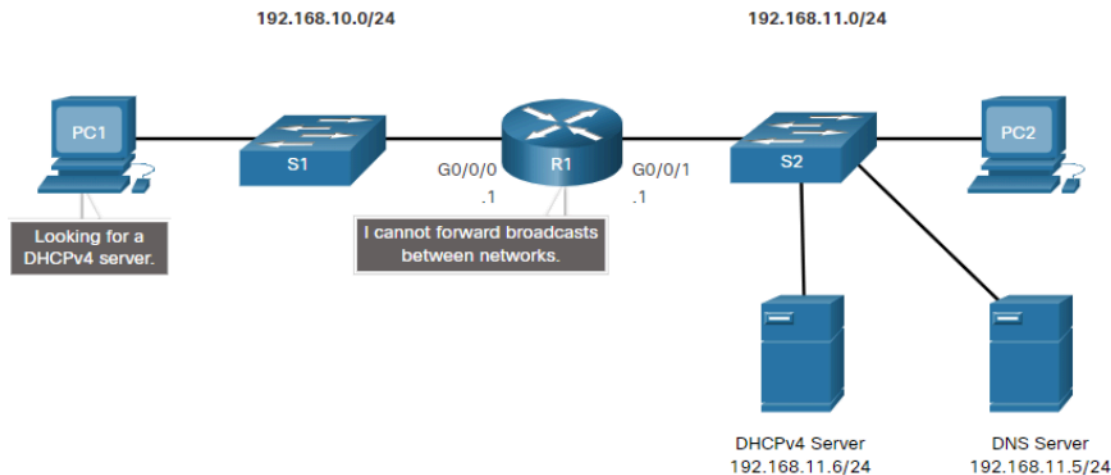
Note: Clearing the DHCP bindings or stopping and restarting the DHCP service may result in duplicate IP addresses being temporarily assigned on the network.

```
R1(config)# no service dhcp
R1(config)# service dhcp
R1(config)#
```

Configure a Cisco IOS DHCPv4 Server

DHCPv4 Relay

- In a complex hierarchical network, enterprise servers are usually located centrally. These servers may provide DHCP, DNS, TFTP, and FTP services for the network. Network clients are not typically on the same subnet as those servers. In order to locate the servers and receive services, clients often use broadcast messages.
- In the figure, PC1 is attempting to acquire an IPv4 address from a DHCPv4 server using a broadcast message. In this scenario, R1 is not configured as a DHCPv4 server and does not forward the broadcast. Because the DHCPv4 server is located on a different network, PC1 cannot receive an IP address using DHCP. R1 must be configured to relay DHCPv4 messages to the DHCPv4 server.



Configure a Cisco IOS DHCPv4 Server

DHCPv4 Relay (Cont.)

- Configure R1 with the **ip helper-address address** interface configuration command. This will cause R1 to relay DHCPv4 broadcasts to the DHCPv4 server. As shown in the example, the interface on R1 receiving the broadcast from PC1 is configured to relay DHCPv4 address to the DHCPv4 server at 192.168.11.6.
- When R1 has been configured as a DHCPv4 relay agent, it accepts broadcast requests for the DHCPv4 service and then forwards those requests as a unicast to the IPv4 address 192.168.11.6. The network administrator can use the **show ip interface** command to verify the configuration.

```
R1(config)# interface g0/0/0
R1(config-if)# ip helper-address 192.168.11.6
R1(config-if)# end
R1#
```

```
R1# show ip interface g0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
Internet address is 192.168.10.1/24
Broadcast address is 255.255.255.255
Address determined by setup command
MTU is 1500 bytes
Helper address is 192.168.11.6
(output omitted)
```

Configure a Cisco IOS DHCPv4 Server

Other Service Broadcasts Relayed

DHCPv4 is not the only service that the router can be configured to relay. By default, the **ip helper-address** command forwards the following eight UDP services:

- Port 37: Time
- Port 49: TACACS
- Port 53: DNS
- Port 67: DHCP/BOOTP server
- Port 68: DHCP/BOOTP client
- Port 69: TFTP
- Port 137: NetBIOS name service
- Port 138: NetBIOS datagram service

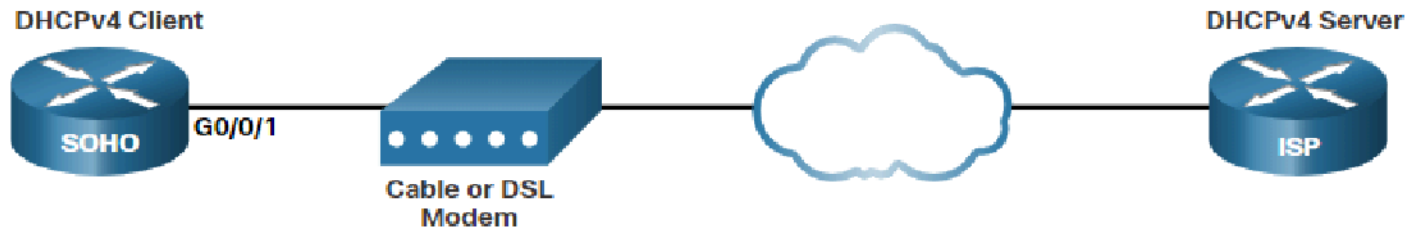
7.3 Configure a DHCPv4 Client

Configure a DHCPv4 Client

Cisco Router as a DHCPv4 Client

There are scenarios where you might have access to a DHCP server through your ISP. In these instances, you can configure a Cisco IOS router as a DHCPv4 client.

- Sometimes, Cisco routers in a small office or home office (SOHO) and branch sites have to be configured as DHCPv4 clients in a similar manner to client computers. The method used depends on the ISP. However, in its simplest configuration, the Ethernet interface is used to connect to a cable or DSL modem.
- To configure an Ethernet interface as a DHCP client, use the **ip address dhcp interface** configuration mode command.
- In the figure, assume that an ISP has been configured to provide select customers with IP addresses from the 209.165.201.0/27 network range after the G0/0/1 interface is configured with the **ip address dhcp** command.



Configure a DHCPv4 Client

Configuration Example

- To configure an Ethernet interface as a DHCP client, use the **ip address dhcp** interface configuration mode command, as shown in the example. This configuration assumes that the ISP has been configured to provide select customers with IPv4 addressing information.
- The **show ip interface g0/1** command confirms that the interface is up and that the address was allocated by a DHCPv4 server.

```
SOHO(config)# interface G0/0/1
SOHO(config-if)# ip address dhcp
SOHO(config-if)# no shutdown
Sep 12 10:01:25.773: %DHCP-6-ADDRESS_ASSIGN: Interface GigabitEthernet0/0/1 assigned DHCP address
209.165.201.12, mask 255.255.255.224, hostname SOHO
```

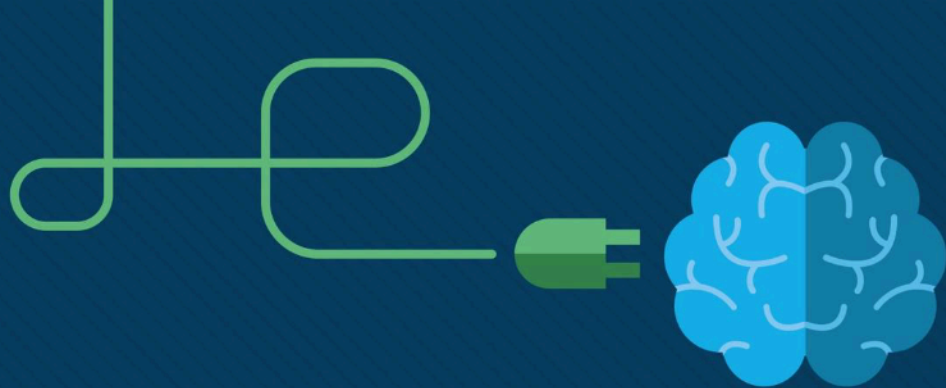
```
SOHO# show ip interface g0/0/1
GigabitEthernet0/0/1 is up, line protocol is up
  Internet address is 209.165.201.12/27
  Broadcast address is 255.255.255.255
  Address determined by DHCP
(output omitted)
```


Configure a DHCPv4 Client Home Router as a DHCPv4 Client

Home routers are typically already set to receive IPv4 addressing information automatically from the ISP. This is so that customers can easily set up the router and connect to the internet.

- For example, the figure shows the default WAN setup page for a Packet Tracer wireless router. Notice that the internet connection type is set to **Automatic Configuration - DHCP**. This selection is used when the router is connected to a DSL or cable modem and acts as a DHCPv4 client, requesting an IPv4 address from the ISP.
- Various manufacturers of home routers will have a similar setup.

The screenshot displays the configuration interface for a 'Wireless Tri-Band Home Router'. At the top, the title 'Wireless Tri-Band Home Router' is on the left, and 'Firmware Version: v0.9.7' is on the right. Below this is a navigation bar with tabs: 'Setup' (selected), 'Wireless', 'Security', 'Access Restrictions', 'Applications & Gaming', 'Administration', and 'Status'. Under the 'Setup' tab, there are sub-tabs: 'Basic Setup', 'OCNs', 'MAC Address Clone', and 'Advanced Routing'. The main content area is titled 'Internet Setup'. It features a dropdown menu for 'Internet Connection type' set to 'Automatic Configuration - DHCP'. Below this, there are input fields for 'Host Name', 'Domain Name', and 'MTU' (with a dropdown arrow) and 'Size' (set to 1500). A 'Help...' link is located on the right side of the 'Internet Setup' section.



Module 8: SLAAC and DHCPv6

Switching, Routing and Wireless
Essentials v7.0 (SRWE)



Module Objectives

Module Title: SLAAC and DHCPv6

Module Objective: Configure dynamic address allocation in IPv6 networks.

Topic Title	Topic Objective
IPv6 Global Unicast Address Assignment	Explain how an IPv6 host can acquire its IPv6 configuration.
SLAAC	Explain the operation of SLAAC.
DHCPv6	Explain the operation of DHCPv6
Configure DHCPv6 Server	Configure a stateful and stateless DHCPv6 server.

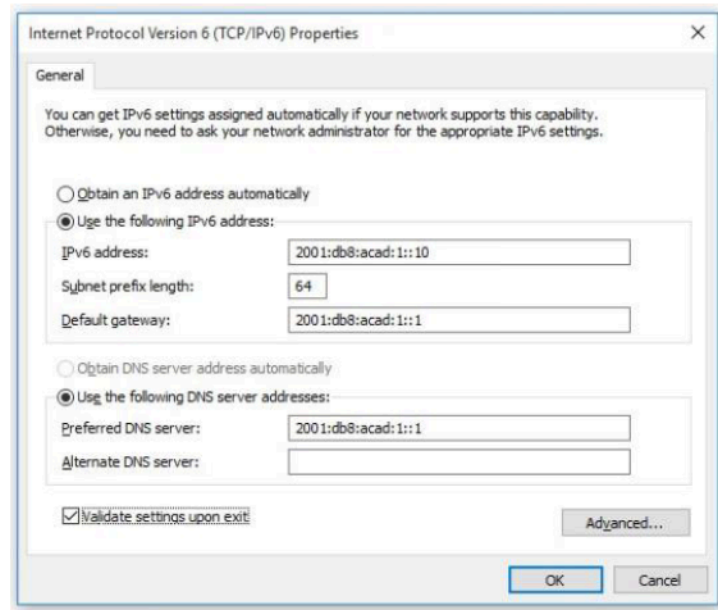
8.1 IPv6 GUA Assignment

IPv6 GUA Assignment

IPv6 Host Configuration

On a router, an IPv6 global unicast address (GUA) is manually configured using the **ipv6 address** *ipv6-address/prefix-length* interface configuration command.

- A Windows host can also be manually configured with an IPv6 GUA address configuration, as shown in the figure.
- However, manually entering an IPv6 GUA can be time consuming and somewhat error prone.
- Therefore, most Windows host are enabled to dynamically acquire an IPv6 GUA configuration.



IPv6 Host Link-Local Address

If automatic IPv6 addressing is selected, the host will use an Internet Control Message Protocol version 6 (ICMPv6) Router Advertisement (RA) message to help it autoconfigure an IPv6 configuration.

- The IPv6 link-local address is automatically created by the host when it boots and the Ethernet interface is active.
- The interface did not create an IPv6 GUA in the output because the network segment did not have a router to provide network configuration instructions for the host.
- **Note:** The "%" and number at the end of the link-local address is known as a Zone ID or Scope ID and is used by the OS to associate the LLA with a specific interface.
- **Note:** DHCPv6 is defined in RFC 3315.

```
C:\PC1> ipconfig
Windows IP Configuration
Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 
    Link-local IPv6 Address . . . . . : fe80::fb:1d54:839f:f595%21
    IPv4 Address. . . . . : 169.254.202.140
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . : 

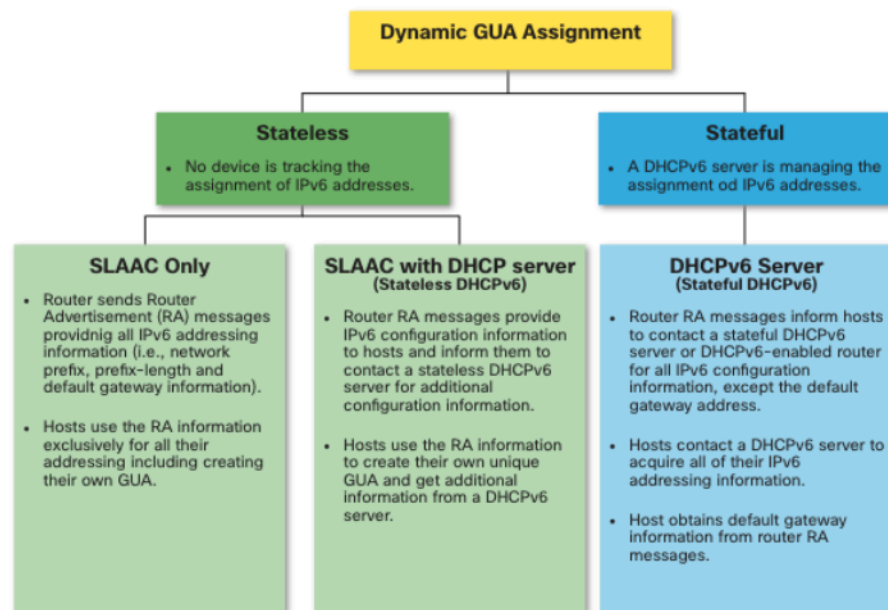
C:\PC1>
```

IPv6 GUA Assignment

IPv6 GUA Assignment

By default, an IPv6-enabled router periodically send ICMPv6 RAs which simplifies how a host can dynamically create or acquire its IPv6 configuration.

- A host can dynamically be assigned a GUA using stateless and stateful services.
- All stateless and stateful methods in this module use ICMPv6 RA messages to suggest to the host how to create or acquire its IPv6 configuration.
- Although host operating systems follow the suggestion of the RA, the actual decision is ultimately up to the host



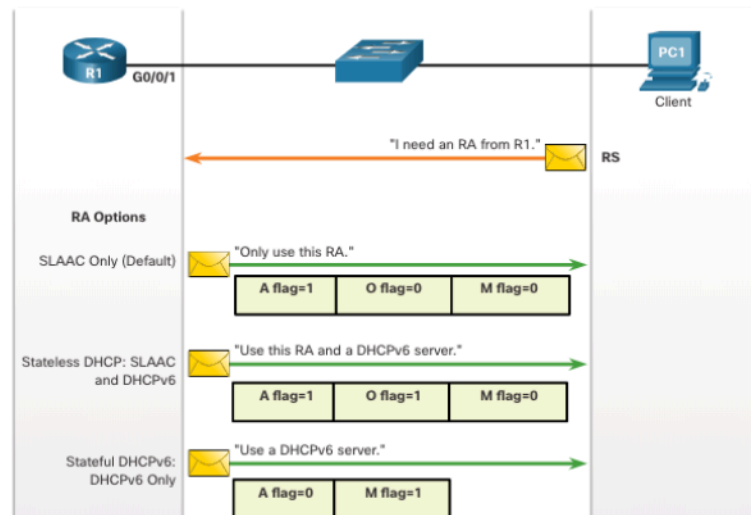
IPv6 GUA Assignment

Three RA Message Flags

How a client obtains an IPv6 GUA depends on settings in the RA message.

An ICMPv6 RA message includes the following three flags:

- **A flag** - The Address Autoconfiguration flag signifies to use Stateless Address Autoconfiguration (SLAAC) to create an IPv6 GUA
- **O flag** - The Other Configuration flag signifies that additional information is available from a stateless DHCPv6 server.
- **M flag** - The Managed Address Configuration flag signifies to use a stateful DHCPv6 server to obtain an IPv6 GUA.



Using different combinations of the A, O and M flags, RA messages inform the host about the dynamic options available.

8.2 SLAAC

SLAAC

SLAAC Overview

Not every network has access to a DHCPv6 server but every device in an IPv6 network needs a GUA. The SLAAC method enables hosts to create their own unique IPv6 global unicast address without the services of a DHCPv6 server.

- SLAAC is a stateless service which means there is no server that maintains network address information to know which IPv6 addresses are being used and which ones are available.
- SLAAC sends periodic ICMPv6 RA messages (i.e., every 200 seconds) providing addressing and other configuration information for hosts to autoconfigure their IPv6 address based on the information in the RA.
- A host can also send a Router Solicitation (RS) message requesting an RA.
- SLAAC can be deployed as SLAAC only, or SLAAC with DHCPv6.

SLAAC

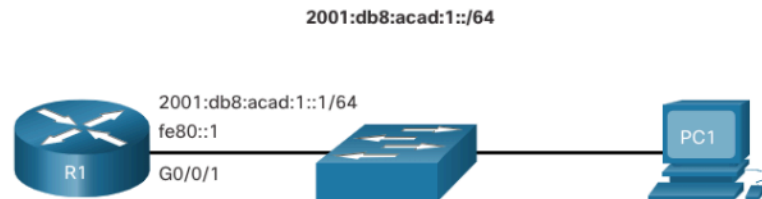
Enabling SLAAC

R1 G0/0/1 has been configured with the indicated IPv6 GUA and link-local addresses.

The R1 G0/0/01 IPv6 addresses include:

- **Link-local IPv6 address** - fe80::1
- **GUA / subnet** - 2001:db8:acad:1::1, 2001:db8:acad:1::/64
- **IPv6 all-nodes group** - ff02::1

R1 is configured to join the all IPv6 multicast group and start sending RA messages containing address configuration information to hosts using SLAAC.



```
R1# show ipv6 interface G0/0/1
GigabitEthernet0/0/1 is up, line protocol is up
  IPv6 is enabled, link-local address is FE80::1
  No Virtual link-local address(es):
  Description: Link to LAN
  Global unicast address(es):
    2001:DB8:ACAD:1::1, subnet is 2001:DB8:ACAD:1::/64
  Joined group address(es):
    FF02::1
    FF02::1:FF00:1
  (output omitted)
R1#
```

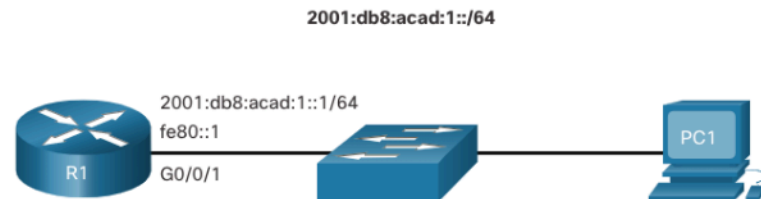
```
R1(config)# ipv6 unicast-routing
R1(config)# exit
R1#
```

SLAAC

Enabling SLAAC (Cont.)

The IPv6 all-routers group responds to the IPv6 multicast address ff02::2.

- The **show ipv6 interface** command verifies that R1 has joined the IPv6 all-routers group (i.e., ff02::2).
- R1 will now begin to send RA messages every 200 seconds to the IPv6 all-nodes multicast address ff02::1.



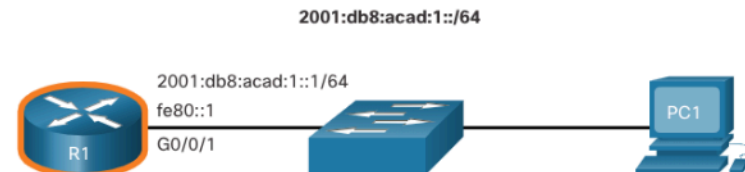
```
R1# show ipv6 interface G0/0/1 | section Joined
Joined group address(es):
  FF02::1
  FF02::2
  FF02::1:FF00:1
R1#
```

SLAAC

SLAAC Only Method

RA messages from R1 have the following flags set:

- **A = 1** – Informs the client to use the IPv6 GUA prefix in the RA and dynamically create its own Interface ID.
- **O = 0** and **M = 0** – Informs the client to also use the additional information in the RA message (i.e., DNS server, MTU, and default gateway information).
- The **ipconfig** Windows command confirms that PC1 has generated an IPv6 GUS using the R1 RA.
- The default gateway address is LLA of the R1 G0/0/1 interface.



RA Message

Flag	value
A	1
O	0
M	0

```
C:\PC1> ipconfig
Windows IP Configuration
Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2001:db8:acad:1:1de9:c69:73ee:ca8c
    Link-local IPv6 Address . . . . . : fe80::fb:1d54:839f:f595%21
    IPv4 Address. . . . . : 169.254.202.140
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . : fe80::1%6

C:\PC1>
```

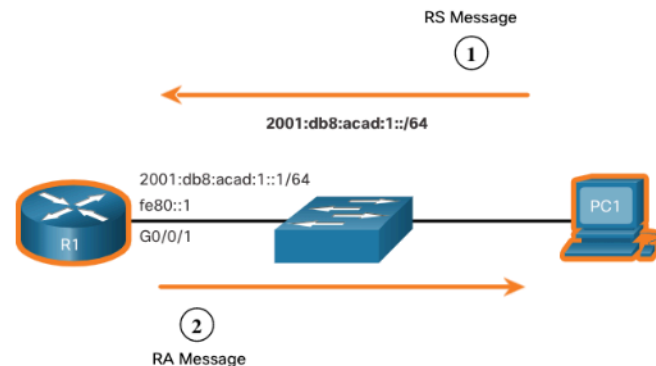
SLAAC ICMPv6 RS Messages

A router sends RA messages every 200 seconds or when it receives an RS message from a host.

- IPv6 enabled hosts wishing to obtain IPv6 addressing information send an RS message to the IPv6 all-routers multicast address of ff02::2.

The figure illustrates how a host initiates the SLAAC method.

1. PC1 has just booted and sends an RS message to the IPv6 all-routers multicast address of ff02::2 requesting an RA.
2. R1 generates an RA and then sends the RA message to the IPv6 all-nodes multicast address of ff02::1. PC1 uses this information to create a unique IPv6 GUA.



Host Process to Generate Interface ID

Using SLAAC, a host acquires its 64-bit IPv6 subnet information from the router RA and must generate the remainder 64-bit interface identifier (ID) using either:

- **Randomly generated** - The 64-bit interface ID is randomly generated by the client operating system. This is the method now used by Windows 10 hosts.
- **EUI-64** - The host creates an interface ID using its 48-bit MAC address and inserts the hex value of fffe in the middle of the address. Some operating systems default to the randomly generated interface ID instead of the EUI-64 method, due to privacy concerns. This is because the Ethernet MAC address of the host is used by EUI-64 to create the interface ID.

Note: Windows, Linux, and Mac OS allow for the user to modify the generation of the interface ID to be either randomly generated or to use EUI-64.

SLAAC Duplicate Address Detection

A SLAAC host may use the following Duplicate Address Detection (DAD) process to ensure that the IPv6 GUA is unique.

- The host sends an ICMPv6 Neighbor Solicitation (NS) message with a specially constructed solicited-node multicast address containing the last 24 bits of IPv6 address of the host.
- If no other devices respond with a Neighbor Advertisement (NA) message, then the address is virtually guaranteed to be unique and can be used by the host.
- If an NA is received by the host, then the address is not unique, and the host must generate a new interface ID to use.

Note: DAD is really not required because a 64-bit interface ID provides 18 quintillion possibilities. Therefore, the chance of a duplicate address is remote. However, the Internet Engineering Task Force (IETF) recommends that DAD is used. Therefore, most operating systems perform DAD on all IPv6 unicast addresses, regardless of how the address is configured.

8.3 DHCPv6

DHCPv6

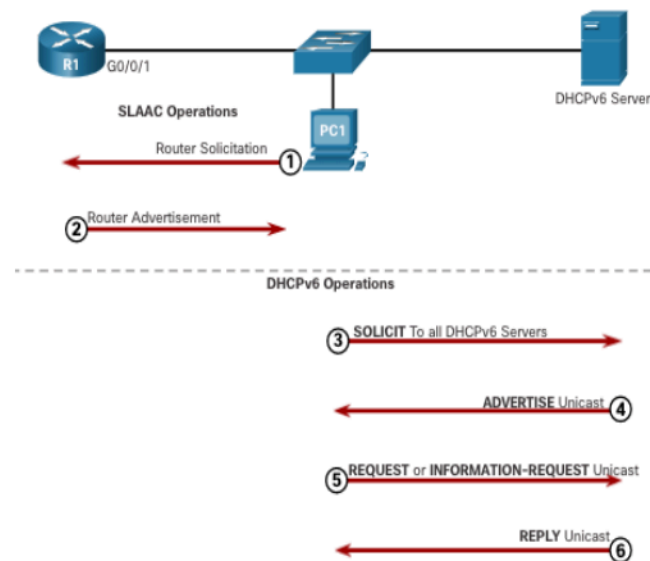
DHCPv6 Operation Steps

Stateful DHCPv6 does not require SLAAC while stateless DHCPv6 does.

Regardless, when an RA indicates to use DHCPv6 or stateful DHCPv6:

1. The host sends an RS message.
2. The router responds with an RA message.
3. The host sends a DHCPv6 SOLICIT message.
4. The DHCPv6 server responds with an ADVERTISE message.
5. The host responds to the DHCPv6 server.
6. The DHCPv6 server sends a REPLY message.

Note: Server to client DHCPv6 messages use UDP destination port 546 while client to server DHCPv6 messages use UDP destination port 547.



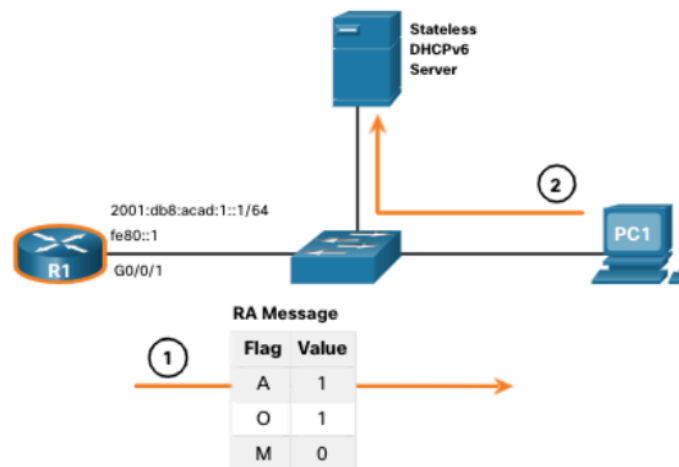
Stateless DHCPv6 Operation

If an RA indicates the stateless DHCPv6 method, the host uses the information in the RA message for addressing and contacts a DHCPv6 server for additional information.

Note: The DHCPv6 server only provides configuration parameters for clients and does not maintain a list of IPv6 address bindings (i.e. stateless).

For example, PC1 receives a stateless RA message containing:

- The IPv6 GUA network prefix and prefix length.
 - A flag set to 1 informing the host to use SLAAC.
 - O flag set to 1 informing the host to seek that additional configuration information from a DHCPv6 server.
 - M flag set to the default value 0.
-
- PC1 sends a DHCPv6 SOLICIT message seeking additional information from a stateless DHCPv6 server.



Enable Stateless DHCPv6 on an Interface

Stateless DHCPv6 is enabled using the **ipv6 nd other-config-flag** interface configuration command setting the O flag to 1.

The highlighted output confirms the RA will tell receiving hosts to use stateless autoconfigure (A flag = 1) and contact a DHCPv6 server to obtain another configuration information (O flag = 1).

Note: You can use the **no ipv6 nd other-config-flag** to reset the interface to the default SLAAC only option (O flag = 0).

```
R1(config-if)# ipv6 nd other-config-flag
R1(config-if)# end
R1#
R1# show ipv6 interface g0/0/1 | begin ND
  ND DAD is enabled, number of DAD attempts: 1
  ND reachable time is 30000 milliseconds (using 30000)
  ND advertised reachable time is 0 (unspecified)
  ND advertised retransmit interval is 0 (unspecified)
  ND router advertisements are sent every 200 seconds
  ND router advertisements live for 1800 seconds
  ND advertised default router preference is Medium
  Hosts use stateless autoconfig for addresses.
  Hosts use DHCP to obtain other configuration.
R1#
```

DHCPv6

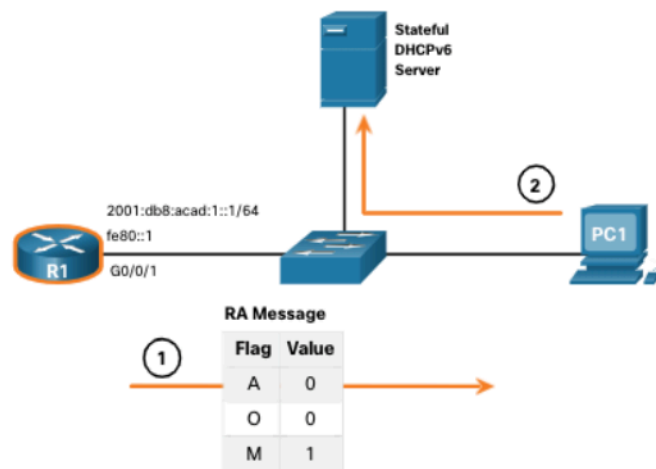
Stateful DHCPv6 Operation

If an RA indicates the stateful DHCPv6 method, the host contacts a DHCPv6 server for all configuration information.

- **Note:** The DHCPv6 server is stateful and maintains a list of IPv6 address bindings.

For example, PC1 receives a stateful RA message containing:

- The IPv6 GUA network prefix and prefix length.
 - A flag set to 0 informing the host to contact a DHCPv6 server.
 - O flag set to 0 informing the host to contact a DHCPv6 server.
 - M flag set to the value 1.
-
- PC1 sends a DHCPv6 SOLICIT message seeking additional information from a stateful DHCPv6 server.



Enable Stateful DHCPv6 on an Interface

Stateful DHCPv6 is enabled using the **ipv6 nd managed-config-flag** interface configuration command setting the M flag to 1.

The highlighted output in the example confirms that the RA will tell the host to obtain all IPv6 configuration information from a DHCPv6 server (M flag = 1).

```
R1(config)# int g0/0/1
R1(config-if)# ipv6 nd managed-config-flag
R1(config-if)# end
R1#
R1# show ipv6 interface g0/0/1 | begin ND
    ND DAD is enabled, number of DAD attempts: 1
    ND reachable time is 30000 milliseconds (using 30000)
    ND advertised reachable time is 0 (unspecified)
    ND advertised retransmit interval is 0 (unspecified)
    ND router advertisements are sent every 200 seconds
    ND router advertisements live for 1800 seconds
    ND advertised default router preference is Medium
    Hosts use DHCP to obtain routable addresses.
R1#
```

8.4 Configure DHCPv6 Server

Configure DHCPv6 Server

DHCPv6 Router Roles

Cisco IOS routers are powerful devices. In smaller networks, you do not have to have separate devices to have a DHCPv6 server, client, or relay agent. A Cisco IOS router can be configured to provide DHCPv6 server services.

Specifically, it can be configured to be one of the following:

- **DHCPv6 Server** - Router provides stateless or stateful DHCPv6 services.
- **DHCPv6 Client** - Router interface acquires an IPv6 IP configuration from a DHCPv6 server.
- **DHCPv6 Relay Agent** - Router provides DHCPv6 forwarding services when the client and the server are located on different networks.

Configure a Stateless DHCPv6 Server

The stateless DHCPv6 server option requires that the router advertise the IPv6 network addressing information in RA messages.

There are five steps to configure and verify a router as a stateless DHCPv6 server:

1. Enable IPv6 routing using the **ipv6 unicast-routing** command.
2. Define a DHCPv6 pool name using the **ipv6 dhcp pool POOL-NAME** global config command.
3. Configure the DHCPv6 pool with options. Common options include **dns-server X:X:X:X:X:X:X:X** and **domain-name name**.
4. Bind the interface to the pool using the **ipv6 dhcp server POOL-NAME** interface config command.
 - Manually change the O flag from 0 to 1 using the **ipv6 nd other-config-flag** interface command. RA messages sent on this interface indicate that additional information is available from a stateless DHCPv6 server. The A flag is 1 by default, telling clients to use SLAAC to create their own GUA.
5. Verify that the hosts have received IPv6 addressing information using the **ipconfig /all** command.

Configure a Stateless DHCPv6 Client

A router can also be a DHCPv6 client and get an IPv6 configuration from a DHCPv6 server, such as a router functioning as a DHCPv6 server.

1. Enable IPv6 routing using the **ipv6 unicast-routing** command.
2. Configure the client router to create an LLA. An IPv6 link-local address is created on a router interface when a global unicast address is configured, or without a GUA using the **ipv6 enable** interface configuration command. Cisco IOS uses EUI-64 to create the Interface ID.
3. Configure the client router to use SLAAC using the **ipv6 address autoconfig** command.
4. Verify that the client router is assigned a GUA using the **show ipv6 interface brief** command.
5. Verify that the client router received other necessary DHCPv6 information. The **show ipv6 dhcp interface g0/0/1** command confirms DHCP option information, such as DNS server and domain name, have been received by the client.

Configure a Stateful DHCPv6 Server

The stateful DHCP server option requires that the IPv6 enabled router tells the host to contact a DHCPv6 server to obtain all necessary IPv6 network addressing information.

There are five steps to configure and verify a router as a stateful DHCPv6 server:

1. Enable IPv6 routing using the **ipv6 unicast-routing** command.
2. Define a DHCPv6 pool name using the **ipv6 dhcp pool POOL-NAME** global config command.
3. Configure the DHCPv6 pool with options. Common options include the **address prefix** command, domain name, DNS server IP address, and more.
4. Bind the interface to the pool using the **ipv6 dhcp server POOL-NAME** interface config command.
 - Manually change the M flag from 0 to 1 using the interface command **ipv6 nd managed-config-flag**.
 - Manually change the A flag from 1 to 0 using the **ipv6 nd prefix default no-autoconfig** interface command to inform the client to not to use SLAAC to create a GUA. The router will now respond to stateful DHCPv6 requests with the information contained in the pool.
5. Verify that the hosts have received IPv6 addressing information using the **ipconfig /all** command.

Configure a Stateful DHCPv6 Client

A router can also be a DHCPv6 client. The client router needs to have **ipv6 unicast-routing** enabled and an IPv6 link-local address to send and receive IPv6 messages.

There are five steps to configure and verify a router as a stateless DHCPv6 client.

1. Enable IPv6 routing using the **ipv6 unicast-routing** command.
2. Configure the client router to create an LLA. An IPv6 link-local address is created on a router interface when a global unicast address is configured, or without a GUA using the **ipv6 enable** interface configuration command. Cisco IOS uses EUI-64 to create an Interface ID.
3. Configure the client router to use DHCPv6 using the **ipv6 address dhcp** interface config command.
4. Verify that the client router is assigned a GUA using the **show ipv6 interface brief** command.
5. Verify that the client router received other necessary DHCPv6 information using the **show ipv6 dhcp interface g0/0/1** command.

DHCPv6 Server Verification Commands

The **show ipv6 dhcp pool** command verifies the name of the DHCPv6 pool and its parameters. The command also identifies the number of active clients.

```
R1# show ipv6 dhcp pool
DHCPv6 pool: IPV6-STATEFUL
  Address allocation prefix: 2001:DB8:ACAD:1::/64 valid 172800 preferred 86400 (2 in use, 0
conflicts)
  DNS server: 2001:4860:4860::8888
  Domain name: example.com
  Active clients: 2
R1#
```

DHCPv6 Server Verification Commands (Cont.)

Use the **show ipv6 dhcp binding** command output to display the IPv6 link-local address of the client and the global unicast address assigned by the server.

- This information is maintained by a stateful DHCPv6 server.
- A stateless DHCPv6 server would not maintain this information.

```
R1# show ipv6 dhcp binding
Client: FE80::192F:6FBC:9DB:B749
  DUID: 0001000125148183005056B327D6
  Username : unassigned
  VRF : default
  IA NA: IA ID 0x03000C29, T1 43200, T2 69120
    Address: 2001:DB8:ACAD:1:A43C:FD28:9D79:9E42
              preferred lifetime 86400, valid lifetime 172800
              expires at Sep 27 2019 09:10 AM (171192 seconds)
Client: FE80::2FC:BAFF:FE94:29B1
  DUID: 0003000100FCBA9429B0
  Username : unassigned
  VRF : default
  IA NA: IA ID 0x00060001, T1 43200, T2 69120
    Address: 2001:DB8:ACAD:1:B4CB:25FA:3C9:747C
              preferred lifetime 86400, valid lifetime 172800
              expires at Sep 27 2019 09:29 AM (172339 seconds)

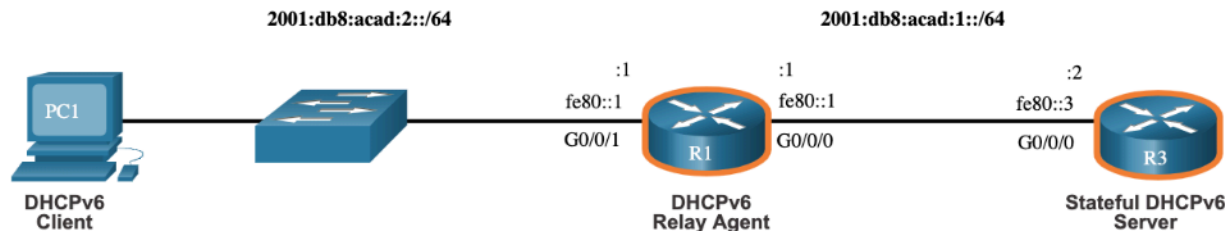
R1#
```

Configure DHCPv6 Server

Configure a DHCPv6 Relay Agent

If the DHCPv6 server is located on a different network than the client, then the IPv6 router can be configured as a DHCPv6 relay agent.

- The configuration of a DHCPv6 relay agent is similar to the configuration of an IPv4 router as a DHCPv4 relay.
- This command is configured on the interface facing the DHCPv6 clients and specifies the DHCPv6 server address and egress interface to reach the server, as shown in the output. The egress interface is only required when the next-hop address is an LLA.



```
R1(config)# interface gigabitethernet 0/0/1
R1(config-if)# ipv6 dhcp relay destination 2001:db8:acad:1::2 G0/0/0
R1(config-if)# exit
R1(config)#
```

Configure DHCPv6 Server

Verify the DHCPv6 Relay Agent

Verify that the DHCPv6 relay agent is operational with the **show ipv6 dhcp interface** and **show ipv6 dhcp binding** commands.

```
R1# show ipv6 dhcp interface
GigabitEthernet0/0/1 is in relay mode
  Relay destinations:
    2001:DB8:ACAD:1::2
    2001:DB8:ACAD:1::2 via GigabitEthernet0/0/0
R1#
```

```
R3# show ipv6 dhcp binding
Client: FE80::5C43:EE7C:2959:DA68
  DUID: 0001000124F5CEA2005056B3636D
  Username : unassigned
  VRF : default
  IA NA: IA ID 0x03000C29, T1 43200, T2 69120
    Address: 2001:DB8:ACAD:2:9C3C:64DE:AADA:7857
              preferred lifetime 86400, valid lifetime 172800
              expires at Sep 29 2019 08:26 PM (172710 seconds)
R3#
```

Verify Windows hosts received IPv6 addressing information with the **ipconfig /all** command.

